

Q1) Prepare the note on Insider threats.

Insider Threats:

Definition

An insider threat is a cybersecurity risk that originates from individuals within an organization who have authorized access to systems and data but misuse that access, either intentionally or unintentionally, causing security incidents.

Insider threats are difficult to detect because they come from trusted individuals who already have legitimate access to sensitive information. Unlike external attackers, insiders do not need to bypass security systems, which increases the risk of data theft, system damage, or unauthorized disclosure. Organizations must therefore monitor internal activities carefully and enforce strict access control policies.

Types of Insider Threats

1. Malicious Insider

A malicious insider intentionally abuses their access privileges to steal confidential data, sabotage systems, or leak sensitive information for personal, financial, or political gain.

2. Negligent Insider

A negligent insider does not intend to cause harm but compromises security through careless behavior such as weak passwords, sharing credentials, or falling victim to phishing attacks.

3. Compromised Insider

A compromised insider occurs when an attacker gains unauthorized access to a legitimate user's account and uses that account to perform malicious actions within the organization.

Causes

- Lack of employee awareness about cybersecurity
- Excessive access privileges given to users
- Weak monitoring and logging systems
- Poor security policies and enforcement

Impact

Insider threats can result in serious consequences including data breaches, financial losses, operational disruption, and damage to organizational reputation and customer trust.

Prevention

- Implement least privilege access control
- Conduct regular employee security training
- Monitor user activities and system logs
- Use multi-factor authentication for account protection

Q2) Case Study of a Major Cyber Attack.

Case Study: WannaCry Ransomware Attack (2017)

Background

In May 2017, a large-scale cyber attack known as WannaCry affected organizations across the world. It was a ransomware attack that encrypted user files and demanded payment in cryptocurrency to restore access.

Cause of the Attack

The attack exploited a vulnerability in Microsoft Windows systems known as EternalBlue. Many organizations had not installed the available security patch, which allowed the malware to spread rapidly across networks.

Impact

The attack affected more than 150 countries and infected hundreds of thousands of computers. Hospitals in the United Kingdom were forced to cancel surgeries and shut down systems, causing serious disruption to healthcare services.

Response and Mitigation

Microsoft released emergency patches and cybersecurity teams worked to contain the spread. Organizations began updating their systems and strengthening security measures to prevent similar attacks.

Lessons Learned

The WannaCry incident highlighted the importance of timely software updates, regular system maintenance, and strong network security practices to prevent large-scale cyber attacks.

Q3. Types of Denial of Service (DoS).

Definition

A Denial of Service (DoS) attack is a cyber attack that attempts to make a computer system, network, or online service unavailable to legitimate users by overwhelming it with excessive traffic or malicious requests.

DoS attacks target system resources such as bandwidth, memory, and processing power. When these resources are exhausted, the system becomes slow or stops responding, preventing legitimate users from accessing services.

Types of DoS Attacks

1. Volume-Based Attacks

Volume-based attacks flood the target network with a large amount of data traffic, consuming available bandwidth and preventing legitimate users from accessing the service.

2. Protocol-Based Attacks

Protocol-based attacks exploit weaknesses in network protocols such as TCP or ICMP. These attacks send incomplete or malformed packets that force the server to allocate resources, eventually exhausting them.

3. Application Layer Attacks

Application layer attacks target specific applications or services, such as web servers, by sending repeated requests that appear legitimate but consume excessive server resources, leading to slow performance or crashes.

Distributed Denial of Service (DDoS)

In a Distributed Denial of Service attack, multiple compromised computers, often part of a botnet, simultaneously send traffic to a single target, making the attack more powerful and difficult to defend against.

Impact

- Website and service downtime
- Financial losses for businesses
- Loss of customer trust
- Disruption of online services

Prevention

- Use firewalls and intrusion detection systems
- Implement traffic filtering and rate limiting
- Deploy load balancing to distribute traffic
- Monitor network activity for unusual patterns

Cyber threats such as insider attacks, ransomware incidents like WannaCry, and Denial of Service attacks demonstrate the increasing need for strong cybersecurity practices. Organizations must adopt proactive security measures, regularly update systems, and educate employees to reduce the risk of cyber attacks and ensure the safety of digital assets.